11. Find an example where $P(AB) < P(A)P(B)$.

let event A = pick a boy aged below 10 from a group of boys aged 7, 8, 9, 10, 11

event B: pick a boy aged above 10 from a group of boys aged 7, 8, 9, 10, 11

$AB = \emptyset$, $P(AB) = 0$, $P(A) = \frac{3}{5}$ $P(B) = \frac{1}{5}$ $P(A)P(B) = \frac{3}{25} > P(AB)$

for $P(A) > 0$, $P(B) > 0$, and $AB = \emptyset$, $P(AB) = 0$

$\therefore P(AB) < P(A)P(B)$

12. Prove (2.3.7) by first showing that

$$(A \cup B) - A = B - (A \cap B).$$

LHS: $(A \cup B) - A$

$= (A \cup B) \cap A^c$

$= (A \cap A^c) \cup (B \cap A^c)$

$= \emptyset \cup (B \cap A^c)$

$= (B \cap B^c) \cup (B \cap A^c)$

$= B \cap (A \cap B)^c$

$= B - (A \cap B)$

$=$ RHS (proven)

To prove (2.3.7),

$\because (A \cup B) - A = B - (A \cap B)$

$P[(A \cup B) - A] = P[B - (A \cap B)]$

$P(A \cup B) - P(A) = P(B) - P(A \cap B)$

$P(A \cup B) + P(A \cap B) = P(A) + P(B)$ (proved)

13. Two groups share some members. Suppose that Group A has 123, Group B has 78 members, and the total membership in both groups is 184. How many members belong to both?

$A = 123$

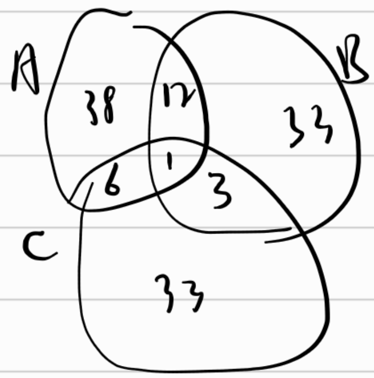
$B = 78$

$A \cap D = ?$

$123 + 78 - x = 184$

$x = 17$

14. Groups A, B, C have 57, 49, 43 members, respectively. A and B have 13, A and C have 7, B and C have 4 members in common; and there is a lone guy who belongs to all three groups. Find the total number of people in all three groups.



$\text{Total} = 38 + 33 + 33 + 12 + 6 + 3 + 1$

$= 126$

22. What is the probability (in the sense of Example 10) that a natural number picked at random is not divisible by any of the numbers 3, 4, 6 but is divisible by 2 or 5?

$A = \{x / x \text{ is a multiple of } 2\}$

$B = \{x / x \text{ is a multiple of } 3\}$

$C = \{x / x \text{ is a multiple of } 4\}$

$D = \{x / x \text{ is a multiple of } 5\}$

$E = \{x / x \text{ is a multiple of } 6\}$

$(B \cup C \cup (AB))^c \cap (A \cup B) = (B \cup C)^c \cap (A \cup D)$

$= B^c C^c (A \cup D)$

$= (B^c C^c A) \cup (B^c C^c D)$

$P = P(B^c C^c A) + P(B^c C^c (AB)) - P(B^c C^c AD)$

$= [P(A) - P(C) + P(ABC) - P(AB))] + \left(\frac{1}{5} - \frac{1}{15} - \frac{1}{20} + \frac{1}{60}\right) - \left(\frac{1}{10} - \frac{1}{30} - \frac{1}{20} + \frac{1}{60}\right)$

$= \frac{7}{30}$

25. Show that if the two events (A, B) are independent, then so are (A, B^c), (A^c, B), and (A^c, B^c). Generalize this result to three independent events.

For (A, B^c)

$A = (AB) \cup (AB^c)$

$P(A) = P(AB) + P(AB^c)$ A and B^c are independent.

$P(A) = P(A)P(B) + P(AB^c)$

$P(AB^c) = P(A)(1 - P(B))$

$P(AB^c) = P(A)P(B^c)$

Similarly,

for (A^c, B) $P(B) = P(AB) + P(A^cB)$

$P(A^cB) = P(B)(1 - P(A))$

$P(A^cB) = P(A^c)P(B)$

for (A^c, B^c) $P(B^c) = P(A^cB^c) + P(AB^c)$

$P(A^cB^c) = P(B^c)(1 - P(A))$

$P(A^cB^c) = P(A^c)P(B^c)$

For 3 independent events,

$P(A \cap B \cap C) = P(A)P(B)P(C)$

$P(A \cap B^c \cap C) = P(A)P(B^c)P(C)$

$P(A \cap B^c \cap C) = P(A)P(C) - P(A)P(B)P(C)$

$= P(A)P(C)(1 - P(B))$

$= P(A)P(C)P(B^c)$

$= P(A)P(B^c)P(C)$

$\Rightarrow P(A \cap B \cap C) = P(A)P(B)P(C)$

28. Suppose five coins are tossed; the outcomes are independent but the probability of heads may be different for different coins. Write the probability of the specific sequence $HHTHT$ and the probability of exactly three heads.

let $P(H_n) = X_n$ $P(T_n) = 1 - X_n$

$HHTHT$, $P(HHTHT) = x_1 \cdot x_2 \cdot y_3 \cdot x_4 \cdot y_5$

$P(3H) = X_1 X_2 X_3 y_4 y_5$

The probability is equal to the total sum of 10 different combinations of X_1, X_2, X_3, $P(T_1) = P(T_2)$

2. Three kinds of shirts are on sale. (a) If two men buy one shirt each, how many possibilities are there? (b) If two shirts are sold, how many possibilities are there?

a) $3 \times 3 = 9$ b) $\binom{3+2-1}{2} = \binom{4}{2} = 6$

3. As in No. 2 make up a good question with 3 shirts and 2 men, to which the answer is 2^3, or $\binom{3+2-1}{3}$.

3 shirts are delivered to 2 men in 2 different packages

it contains 0 to 3 shirt

if shirt is distinguishable $= 2^3$

otherwise $\binom{3+2-1}{3}$

5. How many different initials can be formed with two or three letters of the alphabet? How large must the alphabet be in order that 1 million people can be identified by 3-letter initials?

a) $26^2 + 26^3 = 18252$

b) x = size of alphabet $x^3 = 1 \times 10^6$ $X = 100$

8. In how many ways can four boys and four girls pair off? In how many ways can they stand in a row in alternating gender?

a) $4! = 24$ – boys and girls pair off

b) $2 \times 4! \times 4! = 1152$

12. There are two locks on the door and the keys are among the six different ones you carry in your pocket. In a hurry you dropped one somewhere. What is the probability that you can still open the door? What is the probability that the first two keys you try will open the door?

a) $P(\text{correct key dropped}) = \frac{2}{6}$ $\quad P(\text{correct key present}) = \frac{2}{3}$

b) when the order doesn't matter, $P = \frac{2}{3} \times \frac{1}{4} \times \frac{1}{5} = \frac{1}{30}$

when the order matters

$P = \frac{1}{30} \times 2 = \frac{1}{15}$

*14. Three dice are rolled twice. What is the probability that they show the same numbers (a) if the dice are distinguishable, (b) if they are not. [Hint: divide into cases according to the pattern of the first throw: a pair, a triple, or all different; then match the second throw accordingly.]

a) $\frac{6 \times 6 \times 6}{6^6} = \frac{1}{6^3} = \frac{1}{216}$

b)

16. Four shoes are taken at random from five different pairs. What is the probability that there is at least one pair among them?

$$P = 1 - \frac{\binom{5}{4} \times 2^4}{\binom{10}{4}} = \frac{13}{21}$$